

Artificial Intelligence (AI2002)

Final Exam

Date: May 22nd 2024

Course Instructors

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Total Time (Hrs): 3

Total Marks: 100

Total Questions: 6

Roll No

Section

Student Signature

Attempt all the questions.

CLO # 1: To recognize the notions of rational behavior and intelligent agents.

CLO # 2: To identify and relate methods of search and practically apply the corresponding techniques.

Q1: Answer the following questions briefly.

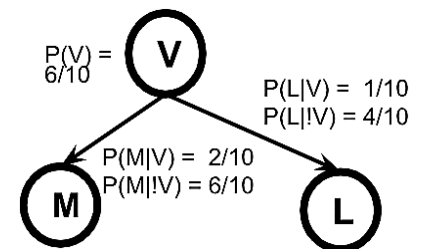
[2x10 = 20 Marks]

- Discuss the importance of problem-solving, knowledge representation, and reasoning in AI systems.
- How do rational agents make decisions to achieve their goals in dynamic and uncertain environments?
- How do different types of environments influence the design and behavior of intelligent agents?
- How can the effectiveness of search algorithms be compared and analyzed?
- Explain how uniform cost search ensures optimality in finding the least-cost path.
- Discuss the challenges posed by repeated states and their impact on search efficiency.
- Discuss the importance of heuristic functions in evaluating search strategies.
- What are sideways moves in hill climbing algorithms and how they contribute to exploration.
- Explain the concept of temperature in simulated annealing and its role in exploration and exploitation.
- How do search strategies differ when solving CSPs compared to path-finding problems?

CLO # 3: To demonstrate understanding and ability to implement the major concepts, approaches and research in probabilistic reasoning, supervised, unsupervised, and other AI areas.

Q2: Pakistan has seen an outbreak of two diseases in cattle, mad cow (M) and lumpy skin (L). A new vaccine has been designed that helps fight both. Given a cow is vaccinated (V) the chances for it to contract either disease is shown in the Bayesian network. since we have no records for cattle, chances that a cow is vaccinated is 6/10.

[4+8+4=16 Marks]



- Find the probability that a cow was administered the new vaccine given it is diagnosed with lumpy skin disease.
- Provided we have no evidence whether a cow is vaccinated or not. Explain and find the probabilities: $P(L|M)$ and $P(V|M)$
- Given the fact that contracting either mad cow or lumpy skin disease builds antibodies to fight the other, therefore, significantly reducing the chances of getting the other disease. How can this fact be incorporated in the given Bayes network? Show the new network and mention the number of parameters required to define this network.

Q3: Given the following $\langle X, Y \rangle$ pairs, simulate the K-means algorithm to identify two clusters in the data.

Point #	1	2	3	4	5	6	7	8	9	10
X	19	17	23	23	11	50	57	22	47	31
Y	9	8	16	20	21	30	38	34	36	50

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Suppose, initially we assign point#1 and 10 as the center of each cluster, respectively. Given that the distance function is Euclidean distance, then do the following: **[4+8+4 = 16 Marks]**

- Simulate the K-means algorithm for one iteration. What are the cluster assignments after one iteration?
- What are the cluster assignments until convergence?
- Make a table for all data points showing their cluster assignments after one iteration and after convergence.

Q4: A Speech Recognition system is trained to recognize five languages: English, Spanish, French, Mandarin, and Arabic. After evaluation, the system's performance is assessed using a confusion matrix. The confusion matrix is given below where rows identify the actual data while columns are for predicted classes:

	English	Spanish	French	Mandarin	Arabic
English	150	10	5	3	2
Spanish	12	130	8	4	6
French	7	5	140	2	6
Mandarin	4	6	3	145	2
Arabic	3	8	7	5	135

Given the confusion matrix above, answer the following questions:

[4x4 = 16 Marks]

- How many instances of Spanish are correctly classified as Spanish?
- What is the total number of instances that are misclassified?
- Calculate the overall accuracy of the Speech Recognition system.
- What is the precision and recall for French?

Q5: Consider the below table for email classification:

[8+8 = 16 Marks]

Email Subject	Sender	Attachment	Number of Words	Spam
Claim Your Prize Now!	prize@offer.com	Yes	20	Yes
Important Information	admin@company.com	No	15	No
Limited Time Offer	discounts@shop.com	No	12	Yes
Request for Proposal	client@business.com	No	10	No
Urgent Action Required	security@bank.com	Yes	18	Yes

- Using the ID3 algorithm, find the root attribute for building a decision tree to predict whether an email is spam or not.
- A new email has been sent containing the subject "Special Discount", sent from "discounts@shop.com" with an attachment, and contains 22 words. Use this as test data to predict whether the email is spam or not using the decision tree model built with the ID3 algorithm. (Note Email Subject: skip this attribute due to its complexity and for Sender: assume each sender is unique for simplicity).

Q6: A small business wants to predict its monthly sales based on the amount of money spent on advertising. The business has collected data over the past 12 months on advertising expenditure (X in hundreds of dollars) and the corresponding sales (Y in thousands of dollars) as shown below: **[4x4 = 16 Marks]**

Month	1	2	3	4	5	6	7	8	9	10	11	12
X	2	3	5	7	1	6	4	8	3	9	2	10
Y	5	7	8	10	3	9	7	11	6	12	4	13

- Obtain a scatter plot of sales versus advertising expenditure.
- Calculate the linear regression equation and interpret the coefficients.
- Estimate the average sales for a month where \$653 is spent on advertising.
- Discuss what is learned when 'learning the regression model' and what the slope and y-intercept represent in this context.